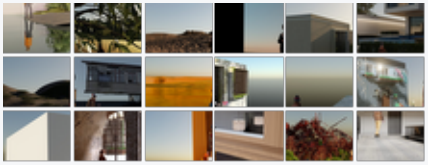
a Assets

BlenderKit



91 indoor & outdoor scenes



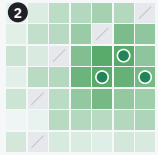
102 human subjects · front calibrated on a 12-view turntable

b VLM-guided placement



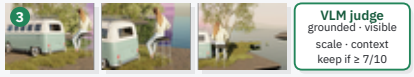
VLM marks where a person belongs

Gemini 2.5 Flash, per scene



ray-cast floor cells

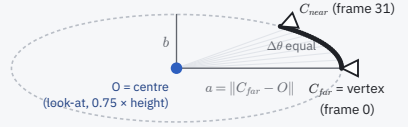
slope ≤ 12°, flat, visible; keep top-3, spread out



place, render up to 6 orbit views; else move & retry

c Camera trajectories

ellipse centred at O with C_far as its vertex; b is solved so it passes through C_near. Endpoints: azimuth ~ U[0°, 360°), distance ~ logU[0.8, 8] m.



rejected (of 393k pairs) — **kept 75k · 19 %**

- 152k subject off frame
- 135k camera blocked
- 17k arc midpoint blocked
- 10k endpoints < 15° apart
- 5k degenerate look-at

d Render & hindsight label

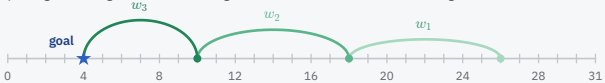


occ 1%	occ 5%	occ 11%	occ 19%	occ 28%	occ 39%	occ 42%	occ 43%	occ 38%
too small	too small	goal-eligible	goal-eligible	goal-eligible	goal-eligible	goal-eligible	goal-eligible	goal-eligible

Each frame is rendered (Cycles, 1024×768) and the subject mesh is projected into it, so its box, bearing, elevation and occupancy are exact — no detector, no annotator. A frame can serve as a goal if occupancy is 10–80 % and ≥ 35 % of the body is visible.

e Windows → action chunks

pick a goal frame g, then starts s with |g − s| ≤ 32; each walks 8 frames toward g



w_3: **9 keyframes (clamped at g)**



a_k: pose change $k \to k{+}1$ in the camera frame, $(\Delta t, R_{6D})$ + shoot, latched once g is reached



8 × 10-D action chunk

≤ 20 windows per trajectory (25 % near goal) · 8 view sectors balanced · 8 val scenes held out

9,889	7,885	393k	75k	2.4M	150k
scene × subject runs	placements (91 scenes)	sampled camera pairs	trajectories (19 % kept)	rendered frames	training windows